CHAPTER

2

Electrochemistry

Sr. No.	Concept	Formulas	Other information
1.	**Cell constant (G*)**	$G^* = \frac{l}{A} cm^{-1}$	Here, l is length and A is cross section area of electrode
2.	**Resistance (R)**	$R = \rho\frac{l}{A} = \frac{1}{\kappa}\frac{l}{A}$ ohm	Here, l is length and A is cross section area of electrode, ρ is resistivity and κ is conductivity.
3.	**Conductance (G)**	$G = \frac{1}{R} ohm^{-1}$ or S or mho $G = \kappa\frac{A}{l}$ ohm^{-1} or S or mho	Here, l is length and A is cross section area of electrode, κ is conductivity. S-seimens
4.	**Conductivity (κ)**	Conductivity = Conductance × Cell constant $\kappa = \frac{1}{\rho} = \frac{1}{R}\left(\frac{l}{A}\right) = G\frac{l}{A}$ $S\ cm^{-1}$ $\kappa = \frac{\text{cell constant}}{R} = \frac{G^*}{R}$	Here, l is length and A is cross section area of electrode, G^* is cell constant and R is resistance
5.	**Resistivity (ρ)**	$\rho = R\frac{A}{l}\ \Omega\ cm$	Here, R is resistance, $\boldsymbol{l}$ is length and A is cross section area of electrode
6.	**Molar conductivity (Λ_m)**	$\Lambda_m = \kappa \times \frac{1000}{C_m} = \kappa \times \frac{1000}{\text{Molarity}}$ $S\ cm^2\ mol^{-1}$	Here, κ is conductivity C_m is molar concentration
7.	**Dissociation constant (K_a)**	$K_a = \frac{C\alpha^2}{(1-\alpha)} = \frac{C\Lambda_m^2}{\Lambda_m^{\infty 2}\left(1-\frac{\Lambda_m}{\Lambda_m^{\infty}}\right)} = \frac{C\Lambda_m^2}{\Lambda_m^{\infty}\left(\Lambda_m^{\infty}-\Lambda_m\right)}$	Here, Λ_m is molar conductivity Λ_m^{∞} is molar conductivity at infinite dilution C is molar concentration α is degree of dissociation.

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8.	**Equivalent Conductance (Λ_{eq})**	$\Lambda_{eq} = \kappa \times \frac{1000}{N} S\,cm^2\,eq^{-1}$	Here, N is normality κ is conductivity
9.	**Kohlrausch's law**	$A_m B_n \rightleftharpoons mA^{n+} + nB^{m-}$ $\Lambda_m^\infty = m\,\Lambda_c^\infty + n\,\Lambda_a^\infty$	Here, Cation $-A^{n+}$ Anion $-B^{m-}$ Λ_m^∞ is molar conductivity at infinite dilution. Λ_c^∞ is molar conductivity of cation at infinite dilution. m and n are Stoichiometric coefficients of cation and anion respectively. Λ_a^∞ is molar conductivity of anion at infinite dilution.
10.	**For strong electrolyte**	$\Lambda_m = \Lambda_m^\infty - A\sqrt{c}$	Here, Λ_m^∞ is molar conductivity at infinite dilution. Λ_m is molar conductivity at any concentration. A – constant c – concentration
11.	**Degree of dissociation(α)**	$\alpha = \frac{\text{Number of molecules ionised}}{\text{Total number of molecules dissolved}}$ $\alpha = \frac{\Lambda_m}{\Lambda_m^\infty}$	Here, Λ_m^∞ is molar conductivity at infinite dilution. Λ_m is molar conductivity at any concentration.
12.	**Electrode potential**	E_{cell} = Reduction potential of cathode – Reduction potential of anode $E^o_{cell} = E^o_{R.P.(Cathode)} + E^o_{O.P.(Anode)}$ $E^o_{cell} = E^o_{R.P.}(\text{R.H.S. electrode}) - E^o_{R.P.}(\text{L.H.S. electrode})$	Here, E_{cell} is cell potential $E^o_{R.P.}$ is standard reduction potential of electrode $E^o_{O.P}$ is standard oxidation potential of electrode E^o_{cell} is standard cell potential

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13.	**Nernst Equation**	$E_{cell} = E^{o}_{cell} - \frac{2.303RT}{nF} \log \frac{[\text{Anode ion}]}{[\text{Cathode ion}]}$ $E_{cell} = E^{o}_{cell} - \frac{0.059}{n} \log K_C$ (At 298 K) $E^{o}_{cell} = \frac{2.303RT}{nF} \log K_C$ (at equilibrium)	Here, E^{o}_{cell} is standard cell potential E_{cell} is the cell potential ΔG^{o} is standard free energy change R is gas constant (8.314 JK^{-1} mol^{-1}). F is Faraday constant (96487 C mol^{-1}) T is temperature in Kelvin n is number of electrons & K_C is equilibrium constant
14.	**Relationship Between ΔG^{o} and Electrode Potential**	$\Delta G^{o} = -nFE^{o}_{cell}$ $\Delta G^{\circ} = -2.303\, RT \log K_C$	Here, n is number of electrons E^{o}_{cell} is standard cell potential ΔG^{o} is standard free energy change F is Faraday constant (96487 C mol^{-1}) K_C is equilibrium constant
15.	**Faraday's Laws of Electrolysis:** (a) First law of electrolysis (b) Second law of electrolysis	 W = ZQ = Z × I × t $Z = \frac{\text{Atomic mass of metal}}{\text{Number of electrons required to reduce the cation}}$ $\frac{W_1}{E_1} = \frac{W_2}{E_2}$	Here, Z is electrochemical equivalent W is weight of substance liberated, I is current (in ampere) and t is time (in second) and Q is amount of charge (in coulomb) W_1 and W_2 are weight of substance liberated at electrode E_1 and E_2 are equivalent weight of substance liberated at electrodes respectively.